

The effect of exposure to alkaline solution and water on the strength–porosity relationship of GFRP rebar

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Received 19 August 2006; accepted 8 November 2007

Available online 22 November 2007

Abstract

This study examined the accelerated degradation of E-glass fiber/vinylester resin composite rebar in alkaline solution and water and analyzed the relationship between the reduction in mechanical performance and the test conditions using microstructural degradation analysis. In this study, a pH of 12.6 was chosen for the alkaline test group, while water was used for the control group. After immersion for up to 300 days, the tensile strength of the glass fiber-reinforced plastic (GFRP) rebar was tested. In addition, the deformation of the GFRP rebar surface was evaluated under scanning electron microscopy (SEM) and the change in the internal pore distribution was measured using mercury intrusion porosimetry (MIP). The mechanical performance decreased as the exposure time and temperature increased. The SEM analysis showed that the internal pore distribution in the GFRP rebar increased as degradation progressed, reducing the mechanical performance. Moreover, the pore distribution increased with temperature, also reducing the mechanical performance. © 2007 Elsevier Ltd. All rights reserved.

Keywords: A. E-glass fiber/vinylester; B. Microstructures; B. Alkaline environment

1. Introduction

To estimate the durability of glass fiber-reinforced plastic (GFRP) rebar, research has focused on the change in rebar under alkaline conditions. Although many studies have examined reinforced materials in alkaline environments, they failed to show the fracture process of GFRP clearly and only suggested that the hydration product at high pH accelerates the destruction of GFRP rebar [1]. The polymer matrix of GFRP rebar resists alkalinity better than reinforced fiber. The matrix not only transfers the load between fibers, but also prevents GFRP rebar from fracturing on exposure to adverse environmental factors [2].

Fig. 1 shows the fracture process of GFRP rebar in alkaline solution. In general, the ultimate deformation of the polymer matrix is greater than that of the fibers, while

the fracture toughness is smaller. When GFRP rebar is stressed, microcracks are generated on the surface of polymer matrix, and the alkaline solution enters the cracks, which ultimately leads to fracture of the rebar [3]. Consequently, when GFRP rebar is used to reinforce concrete, the glass fibers become more brittle because the pore water contains alkaline ions and has a high pH, and the contact layer between the fibers and polymer matrix is damaged as a result of the chemical reaction, increasing the hydration product and reducing the mechanical performance of GFRP rebar.

In most previous studies, GFRP rebar specimens were exposed to artificial concrete pore water at different temperatures and durations. In terms of temperature, most studies examined the range between 20 and 60 °C, although 80 °C was used in some cases. The artificial pore water consisted of NaOH, KOH, and Ca(OH)₂ and had a pH of 12–13.5 [4]. The GFRP specimens were exposed to the alkaline solution or concrete, and then subjected to tensile stress to evaluate the resistance to stress corrosion. Durability was evaluated using the residual strength, elastic

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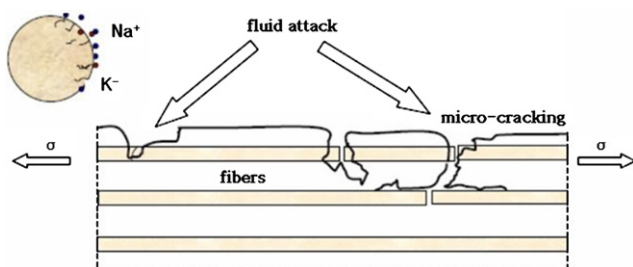


Fig. 1. Degradation of GFRP rebar in alkaline solution.

modulus, and fracture strain. Occasionally, physical analytical equipment and the diffusion test were used to evaluate durability. Normally, the resistance of GFRP rebar to alkalinity was tested using an accelerated test [5].

When GFRP rebar is immersed in solution, water or alkaline is absorbed by the polymer matrix, resulting in delamination and cracks in the fiber matrix. The delamination and cracks act like pores, which allows the further diffusion of water and chemicals and causes degradation. The number and size of the pores increase as the degradation progresses. The delaminations and cracks in the matrix weaken the load transfer between the fibers, and additional harmful chemicals diffuse to the interface between the fiber and matrix, resulting in further degradation of fibers. When water and chemicals reach the fibers through these delaminations or cracks, leaching and etching occur in turn, and the glass fiber surface wears out. Consequently, the interface between the fiber and matrix is weakened, reducing the mechanical performance [6].

Bank et al. immersed GFRP rebar in 0.3%, 3%, and 30% NH_4OH at 23 °C for 224 days to evaluate its durability [12]. They observed a 12% reduction in tensile strength in the 30% solution only. When GFRP rebar was immersed in water at 80 °C, surface cracks were observed.

Uomoto exposed GFRP rebar to $\text{Na}(\text{OH})_2$ solution at 40 °C for 120 days, and found that the tensile strength of the GFRP rebar decreased by 70% [13].

Chin et al. immersed vinylester resin and isotropic polyester resin in 0.68% NaOH and 0.5% $\text{Ca}(\text{OH})_2$ to simulate actual concrete pore water at room temperature; no change in tensile strength or the glass transition temperature occurred after exposure for 1300 h [24].

Buck et al. immersed three different types of GFRP rebar in $\text{Ca}(\text{OH})_2$ solution at 80 °C for 28 days [8]. The tensile strength of each type of rebar was reduced, although the reduction was smallest in the GFRP rebar with a vinylester resin matrix compared to those with matrix containing both vinylester and polyester resins.

Mukherjee and Arwikar prepared solutions at pH 12.25, 12.182, and 12.05 and immersed GFRP rebar for 1 year [9]. The tensile test was repeated every 3 months to measure the mechanical performance reduction and the change in the ion components of fibers, and polymer matrix was monitored using SEM analysis and energy dispersive X-ray analysis (EDX). The strength decreased by about 43% after

3 months and by about 56% after 6 months. The EDX analysis showed that the Si content decreased from 60.93% at baseline to 39.43% after 9 months, quantitatively reflecting the degradation of the glass fibers in the GFRP rebar.

Vistasp et al. immersed rebar in 10.89 and 5.95 g/L CaCO_3 (pH 11.5) at 23, 40, 60, and 80 °C for 80 weeks and measured the mechanical performance reduction through tensile and shear tests; they further observed the degradation of glass fiber in the axial direction and the changes in surface morphology under SEM [10]. They found that the residual tensile strength of the GFRP rebar dropped by 40% after 80 weeks immersion at 80 °C. The residual shear strength was 46%, which occurred because hydrolysis caused by the initial plasticization resulted in marked degradation, as shown in their SEM analysis.

The effect of moisture on GFRP rebar used to reinforce concrete has also been the subject of many studies [14–17]. These have focused on the complex interaction of moisture and heat on the diffusion process and the corresponding mechanical properties of GFRP rebar. Generally, studies of water absorption examine the diffusion process and corresponding mechanical properties based on the complex impact of moisture and heat.

According to Bank et al., when GFRP rebar containing vinylester resin as the polymer matrix is saturated, the water content depends on temperature [1–12]. However, this effect does not become clear unless the GFRP rebar reaches saturation. Furthermore, the fracture of GFRP rebar during degradation is attributable to the increase in the water content due to the increased pores.

As described above, many studies have conducted accelerated degradation tests. However, most of these studies merely evaluated the degradation of GFRP rebar based on the reduction in mechanical performance determined by mechanical tests or observed the surface degradation under SEM [18–25]. This study analyzed the mechanism causing the reduction in mechanical performance of GFRP rebar with accelerated degradation from a microstructural view using SEM and MIP analyses of the change in pore structure.

2. Experimental

2.1. Materials and test method

The GFRP rebar used for the accelerated degradation test was Aslan GFRP rebar, manufactured by Hughes Brothers (USA), which has a tensile strength of 690 MPa, an elastic modulus of 40.8 GPa, and a nominal diameter of 12.7 mm. In order to observe the degradation of the GFRP rebar used as concrete reinforcement, a tensile test was conducted after exposure to the accelerated conditions in accordance with ASTM D 3916, and the performance reduction was evaluated. The setup used for the tensile test is illustrated in Fig. 2. A 250-kN capacity universal testing machine (UTM) for strain control was used to assess the

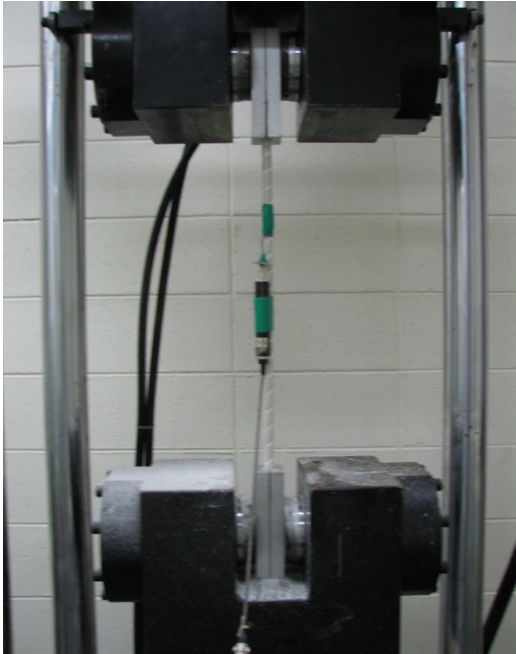


Fig. 2. Setup of the tensile test.

tensile properties. The linear variable displacement transducer (LVDT) was from Tokyo Sokki Kenkyujo and the gage length was 50 mm. The LVDT was attached to the GFRP rebar to measure the strain, as shown in Fig. 2. The test rate used in the GFRP rebar tensile test was 5 mm/min.

In ASTM D 3916, no clear standard exists for the specimen length in the tensile test. Thus, we determined the specimen length for the tensile test in accordance with the ACI committee 440 recommendations [26,27]. The specimen for the tensile test using the ASTM grip had a total length of 80 cm, and its dimensions and shape are shown in Fig. 3.

2.2. Accelerated aging

To evaluate the long-term durability of GFRP rebar, the accelerated test method was used to reduce the degradation time by controlling environmental factors such as temperature. In this study, we controlled temperature to evaluate the durability of GFRP rebar. Control specimens were stored at a constant temperature of 22 °C and relative humidity (RH) of 50% to minimize potential defects result-

ing from moisture absorption and temperature fluctuation and to prevent damage during delivery or cutting, which can affect the mechanical properties. The specimens were exposed to alkaline solution at a pH of 12.6 and water before evaluating their durability.

To simulate the alkaline environment found in real concrete pores, we made a solution of 1.4% KOH, 1% NaOH, and 0.16% $\text{Ca}(\text{OH})_2$. Specimens were immersed in this solution at temperatures of 20, 40, 60, and 80 °C for up to 300 days. As a control, GFRP rebar was immersed in tap water at 20 °C for up to 100 days.

2.3. Microstructure analysis

This study measured the pores using MIP, and then analyzed the degradation of GFRP rebar quantitatively; we observed the surface of the degraded rebar under SEM and then compared the MIP results and SEM observations.

3. Results and discussion

3.1. Tensile test

Fig. 4 plots the residual tensile strength of the GFRP rebar in the alkaline solution as a percentage of the tensile strength of the control group for up to 300 days. The tensile strength decreased as the temperature and exposure increased, especially after 90 days.

The results showed that the tensile strength decreased slightly as the temperature of the alkaline solution increased.

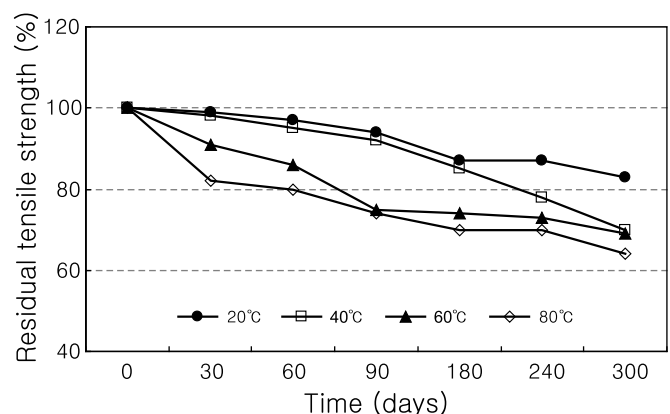


Fig. 4. Residual tensile strength of GFRP rebar after immersion in alkaline solution.

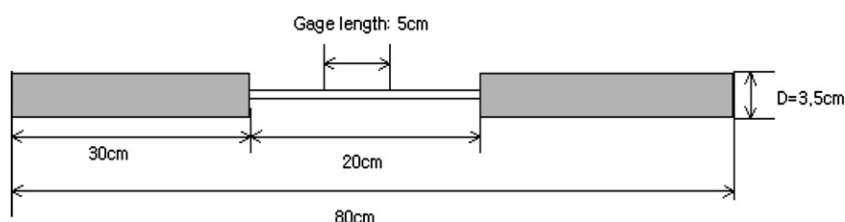


Fig. 3. Dimensions and shape of the GFRP rebar used for the tensile test.

Table 1
Tensile strength of GFRP rebar immersed in tap water

Exposure	Tensile strength (MPa)
50 days	723.43
100 days	708.04

The difference between 20 and 40 °C was not noticeable, while marked differences occurred at 60 and 80 °C.

After immersion for 30 days, the residual tensile strength exceeded 80%, indicating sufficient durability. After 60 days, the results were similar to those at 30 days, although the tensile strength decreased as the temperature increased, especially at 60 and 80 °C. At 60 and 80 °C, the residual tensile strength dropped below 80% after expo-

sure for 90 days. No clear reduction was observed at 20 or 40 °C because the residual strength still exceeded 80%. Exposure to the alkaline solution for 300 days at 20 °C resulted in a residual tensile strength of 83%, while that at 80 °C, the highest accelerating temperature, was 60%. Overall, no marked difference occurred at 20 or 40 °C, while the tensile strength decreased considerably at higher temperatures.

For comparison, the tensile strength of GFRP rebar immersed in normal tap water was measured after 50 and 100 days, and the results are listed in Table 1. The residual tensile strength was 99% and 97%, respectively, which implies that tap water had virtually no effect on the durability.

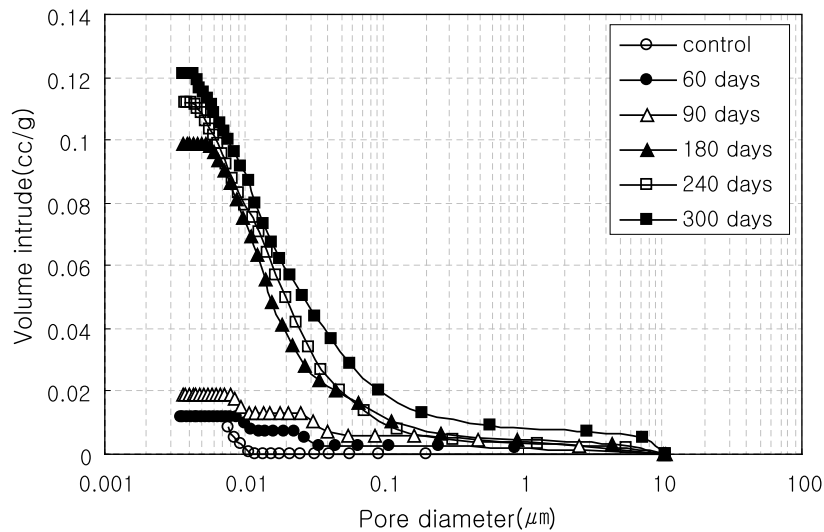


Fig. 5. Change in the pore distribution of GFRP rebar after immersion in alkaline solution at 80 °C for up to 300 days.

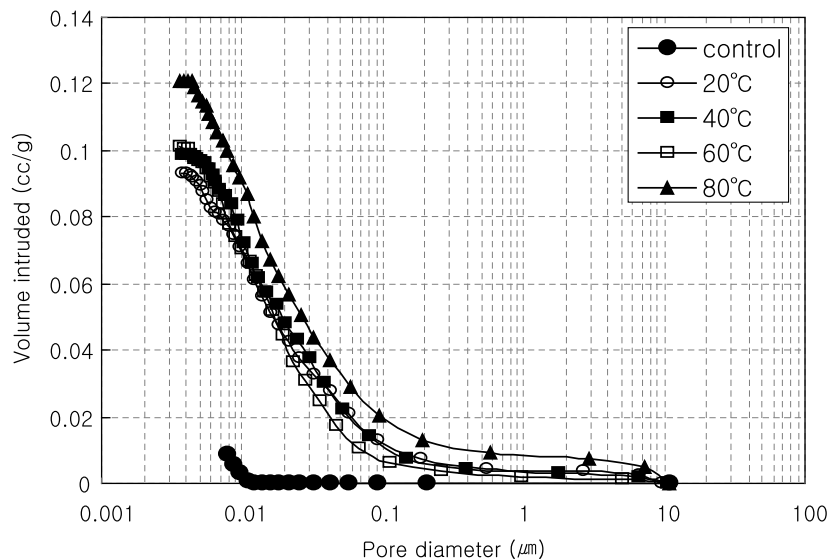


Fig. 6. Change in the pore distribution of GFRP rebar after immersion in alkaline solution at 80 °C.

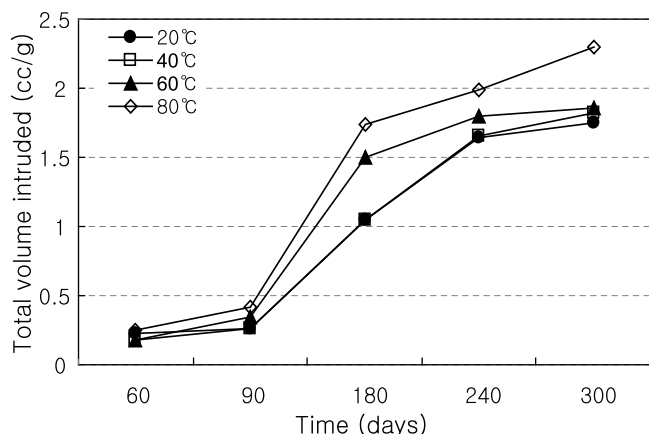


Fig. 7. Change in the total number of pores in GFRP rebar after immersion in alkaline solution for an accelerated duration.

3.2. Microstructure analysis

Fig. 5 plots the pore distribution of GFRP rebar immersed in alkaline solution for different immersion times

at 80 °C. A slight increase occurred in the accumulated mercury indentation after 60 days of immersion, while the mercury indentation increased dramatically after 90 days. The reduction in tensile strength after 90 days took place because matrix delamination and microcracks occurred with degradation of the polymer matrix; the alkaline solution infiltrated the fractures and accelerated the degradation of the GFRP rebar.

Fig. 6 shows the change in pore distribution of GFRP rebar after immersion in alkaline solution for 300 days at different temperatures. The results show that degradation was accelerated at higher temperatures when the other conditions were the same. In other words, higher temperature can accelerate the fracture of GFRP rebar.

Fig. 7 shows that the total volume of pores increased with both immersion time and temperature. Similar to the results for tensile strength, no large difference occurred between 20 and 40 °C. When immersed in alkaline solution for over 90 days, the pores increased dramatically, which reflects the increased delamination and cracks in the polymer matrix of the GFRP rebar or the further reduction in

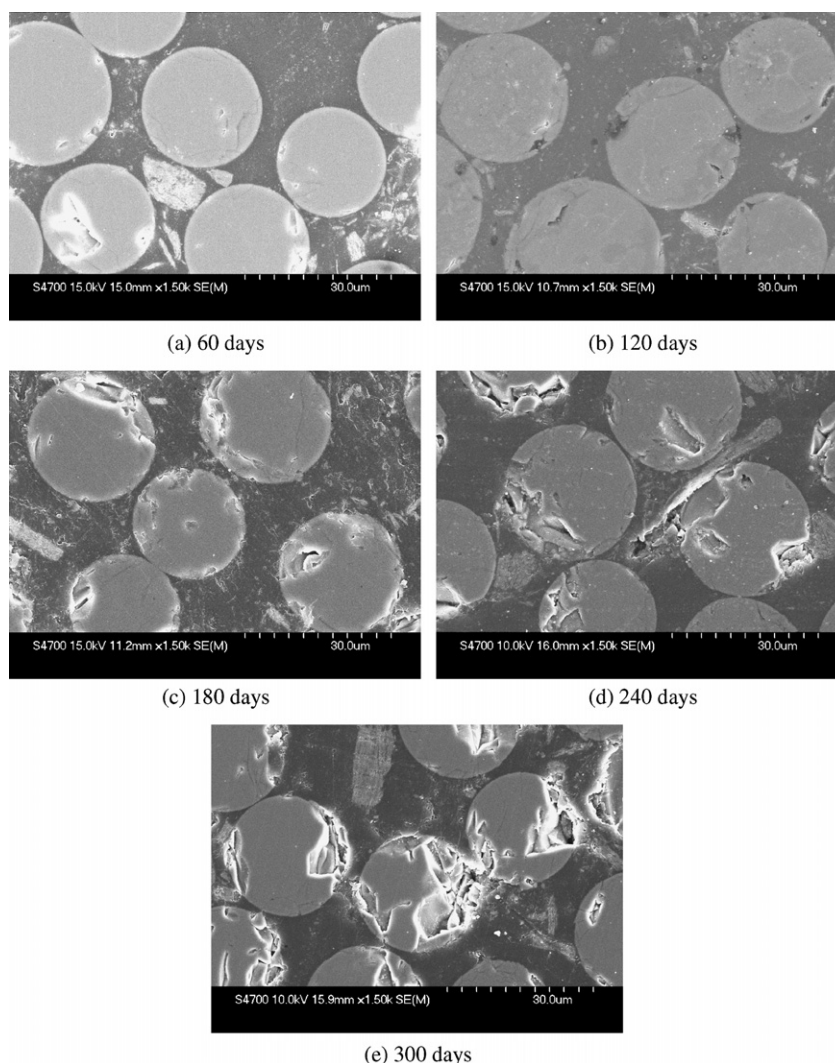


Fig. 8. Surface morphology of GFRP rebar after immersion in alkaline solution at 20 °C.

the cross-sectional area of the glass fibers. When soaked for 180 days, the rates of porosity increase at 80 and 60 °C were 55.9% and 11.4%, respectively, compared to the control specimen. Therefore, the sharp decrease in tensile strength seen after immersion for over 90 days is clearly attributable to the degradation of the GFRP rebar.

Figs. 8–11 show SEM images of the surface morphology of GFRP rebar exposed to alkaline solution at different temperatures for different immersion times. The interface between the fibers and matrix in un-aged GFRP rebar was clear and the fibers were in better condition. At 20 and 40 °C, little surface morphology deformation initially occurred because the immersion temperature was close to normal temperatures; however, degradation took place with increasing immersion time, similar to the change in pore distribution. In contrast, at 60 and 80 °C, the pore volume increased markedly and the surface degradation of the GFRP rebar was accelerated. Water and ions diffused to the interface between the fiber and matrix through the microcracks caused by water absorption.

Ultimately, the signs of degradation became visible as fiber corrosion and a reduction in cross-sectional area. The surface damage to the glass fibers in the alkaline solution increased with the immersion time due to interface degradation, and the increased exposure to the solution at a high pH accelerated the degradation, which ultimately reduced the tensile strength.

Fig. 12 shows the surface morphology of GFRP rebar immersed in tap water for 50 and 100 days. The glass fibers were in far better condition than in an alkaline environment. Although the water absorption by the resin led to swelling of the polymer matrix, which gave rise to delamination and some fiber degradation, the reduction in the cross-sectional area of the glass fibers was not noticeable.

Based on the results of the tensile tests, the reduction in tensile strength became substantial when a specimen was subjected to accelerated degradation for over 90 days at any temperature. As shown in Fig. 5, the reduction in tensile strength at 20 and 40 °C was similar for up to 90 days, but subsequently, the tensile strength of GFRP rebar

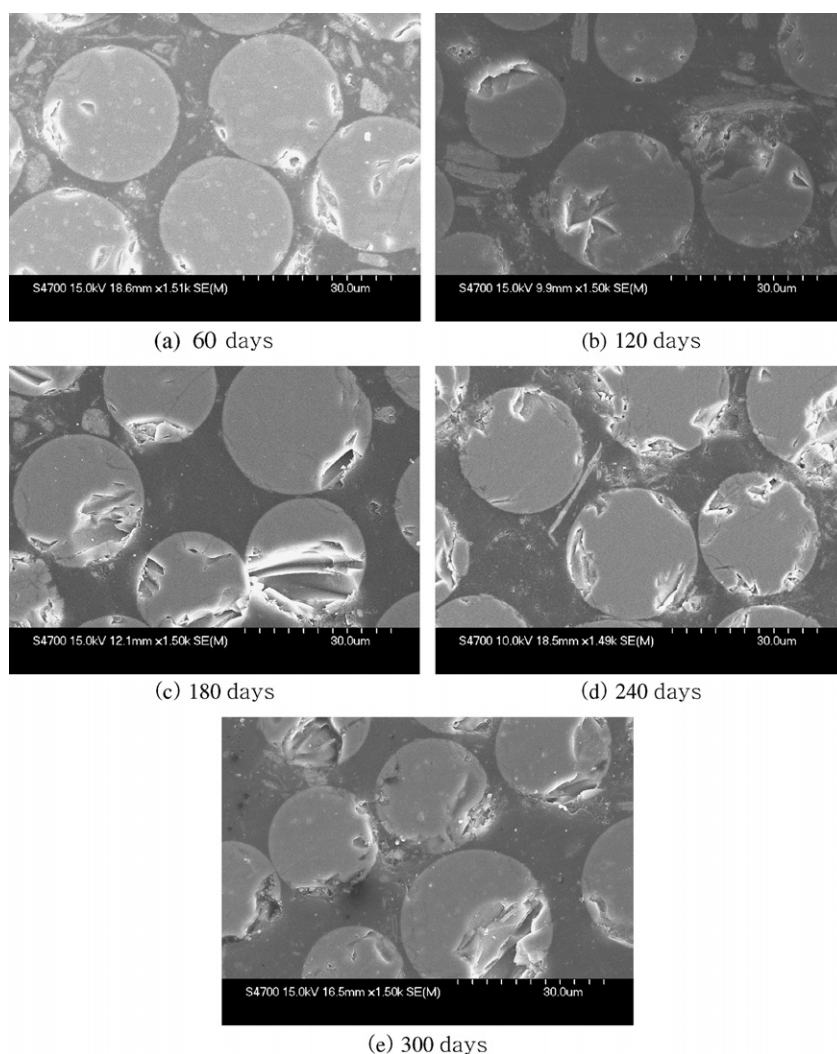


Fig. 9. Surface morphology of GFRP rebar after immersion in alkaline solution at 40 °C.

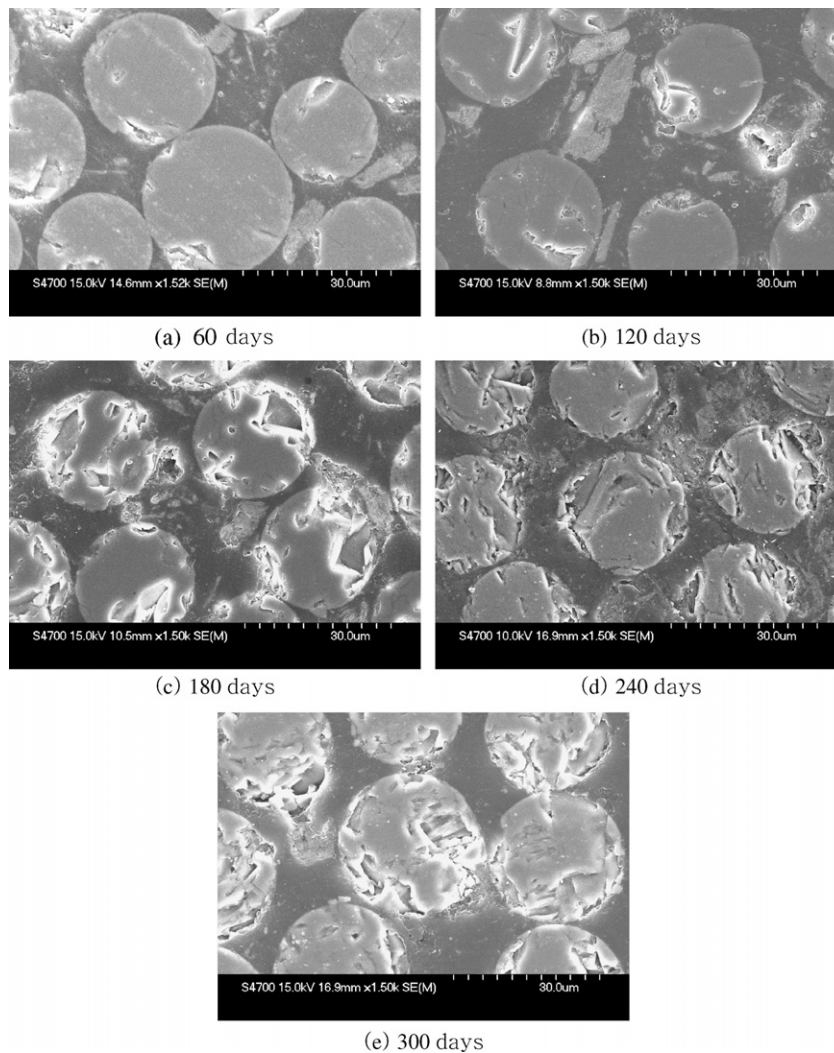


Fig. 10. Surface morphology of GFRP rebar after immersion in alkaline solution at 60 °C.

decreased more rapidly at 40 °C and fell below 80%. At 60 and 80 °C, the residual tensile strength dropped below 80% after 90 days and the degradation then slowed. This was related to the sharp increase in pores, which resulted in delamination and cracks. Consequently, the load transfer performance between fibers weakened, reducing the mechanical performance.

In Figs. 8 and 9, (a) and (b) show the surface morphology of GFRP rebar immersed for 60 and 120 days at 20 and 40 °C, respectively. Since 20 and 40 °C are not very different from normal temperature, few differences were seen under SEM. Even after 60 and 120 days, little obvious degradation was observed that would have influenced the mechanical performance, and the degradation occurred slowly. In contrast, the degradation was marked after subsequent 60-day intervals, as shown in (c–e), and the reduction mechanical performance became major. After 90 days immersion at 60 or 80 °C, the tensile strength was less than 80%, although the reduction tended to level off, as reflected in the changes seen in

(c–e) in Figs. 8 and 9. In Figs. 10 and 11 (a) and (b), the degradation was more intense than in rebar exposed at 20 or 40 °C, but the reduction in the mechanical performance was not likely serious, since many fibers possessed a clear interface; this was confirmed in the tensile tests. However, the increase in fibers with indistinct interface, as shown in (c), and the reduced residual tensile strength below 80% suggest that fiber degradation gave rise to mechanical performance reduction. When exposed at high temperatures, the residual tensile strength fell below 80% at 90 days, although the subsequent reduction was not significant. SEM observations of the surface morphology showed no obvious differences in (d) and (e) compared to (c).

The SEM analysis clarified how the microstructural degradation of GFRP rebar was caused by the increase in pores, which resulted in polymer matrix delamination, cracks, and a reduction in the cross-sectional area of the glass fibers; this contributed to the reduction in the mechanical performance of GFRP rebar.

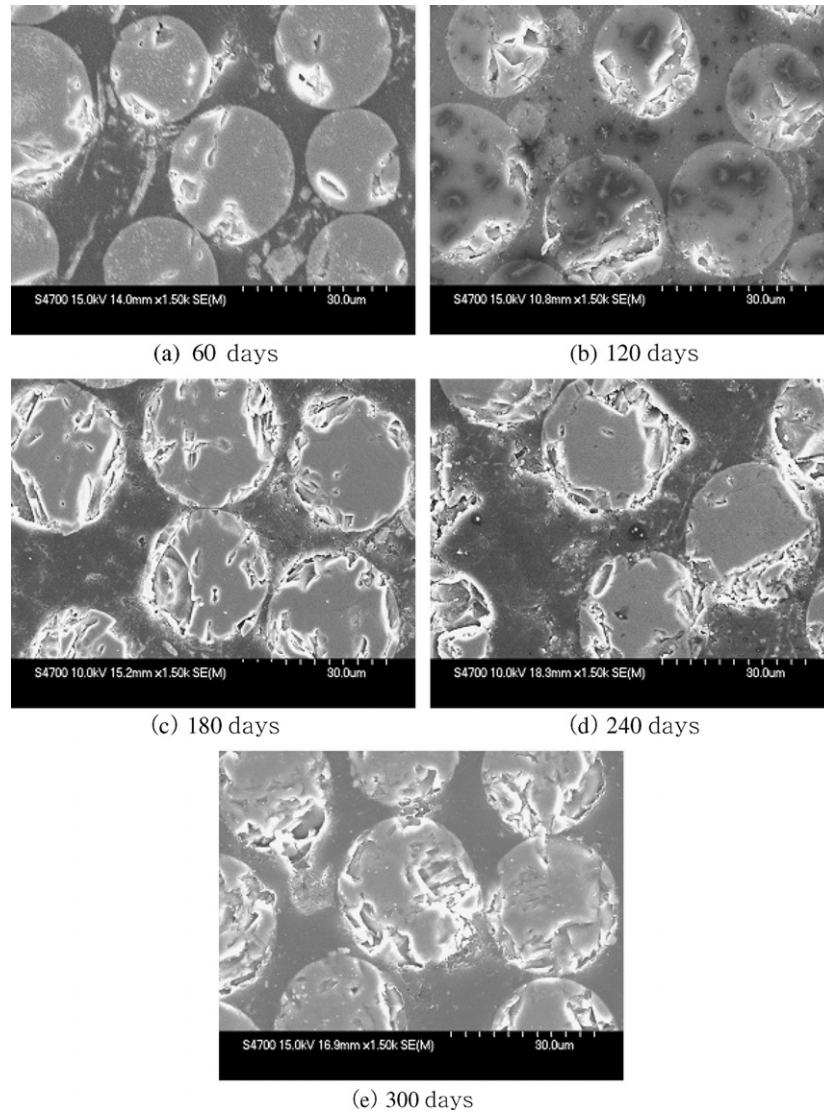


Fig. 11. Surface morphology of GFRP rebar after immersion in alkaline solution at 80 °C.

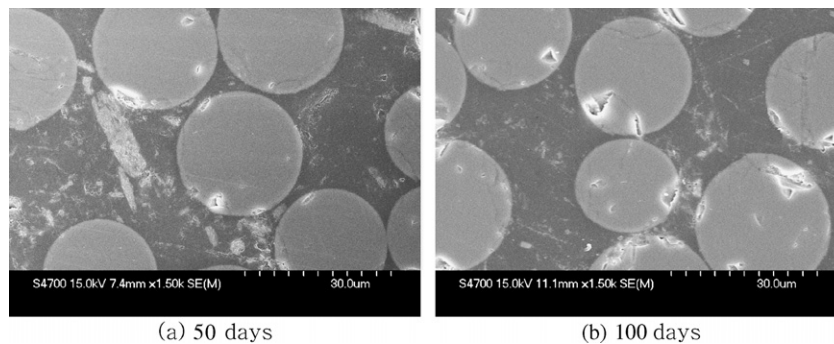


Fig. 12. Surface morphology of GFRP rebar after immersion in tap water.

4. Conclusions

- The tensile strength of GFRP rebar exposed to an alkaline environment decreased markedly as the duration of exposure increased.
- The MIP measurements of the GFRP rebar showed that the gross pore indentation increased from 0.0178 cc/g in the control to 2.3032 cc/g after 300 days immersion in alkaline solution and the accumulated mercury indentation increased with the immersion time. In other words,

when GFRP rebar was exposed to the alkaline solution, the resin absorbed water and plasticization of the polymer matrix began. Consequently, delamination and microcracks were generated and additional water and chemicals diffused through the defects. The increase in pores was attributable to the subsequent degradation of the fiber–matrix interface and surface corrosion. In contrast, the accumulated mercury indentation of GFRP rebar immersed in tap water for 100 days was 0.2061 cc/g, which was negligible compared to exposure to an alkaline environment; the degradation appeared minimal.

- The SEM analysis showed that the degradation of GFRP rebar was accelerated as the exposure to an alkaline environment increased. In contrast, for GFRP rebar immersed in tap water, the reduction in durability was trivial, since the surface degradation was negligible.

Acknowledgements

The work presented in this paper was funded by the Korea Institute of Construction Technology (KICT).

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